

LOADZONE · DESIGN DECISIONS REGISTER

LoadZone design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/loadzone		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work; budget treated as a value-engineering target

LoadZone design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P8 (base tray with locating ring, fill and vents; cradles, standoffs and rib; cells and antenna moved inward; casting tooling; programming hold point; channel brackets with band clamps; screwed display housing with a gland)	Accept; accept with changes; reject	Accept: none changes what LoadZone does, its pitch or its safety case	The whole build plan	LDZ-DDR-003, Table 1
2	First trial partner and LoRaWAN band	A city curb team, a business district or a carrier; EU868 or US915	None yet	Antenna band (BOM line 4); US915 means the 11-byte payload rule	LDZ-DDR-001, 01; LDZ-DDR-002
3	Sealed-for-life puck: cells not replaceable, firmware changed only over the air	(a) accept and plan over-the-air updates for TRL 4; (b) a sealed service port	(a)	Steps 3 to 5; the programming hold point is the last access	LDZ-DDR-003, A1
4	Material of the base tray, which the epoxy bed bonds to	(a) printed ASA, sanded; (b) cast polyurethane in a second mould	(a) for the first prototype, (b) if the TRL 4 pull-off test fails on the tray side	Section 3.5; tooling if (b)	LDZ-DDR-003, A2
5	Fixings showing on the printed face of the sign	(a) button heads; (b) studs bonded to the back of the panel	(a) for the prototype	Sections 3.9 to 3.11	LDZ-DDR-003, A3
6	Sign legend and artwork	Placeholder legend; the local traffic sign rules of the trial city	Set to the trial city's rules once item 2 is decided	Sign face (BOM line 8)	Review note 2026-09-26, item 4
7	Snow and ploughs: surface puck or a cored, in-ground variant where roads are ploughed	Surface puck only; a cored variant for ploughed climates	None yet	Not part of the TRL 3 build	LDZ-PRC-001, open questions; LDZ-DDR-001, D1
8	Dome surface detail in the appearance model (anti-skid grooves, parting line, badge recess cut into the 4 mm wall)	Appearance only; add to the mould master	Appearance only until the mould is designed for production	None in the prototype: the mould master has a plain dome	Review note 2026-09-26, item 2
9	Potting colour shown clear amber in the renders	Render convention; buy a clear grade	Render convention unless a clear grade suits the cost target	Potting (BOM line 6)	Review note 2026-09-26, item 3

#	Decision needed	Options	Recommendation	Affects in the build	Source
	Render layout (kerb and sign drawn closer to the puck) and the host FieldNode left out of the renders	Accept for the renders; redraw to scale	Accept, and mention the FieldNode in the caption	None (renders only)	Review note 2026-09-26, items 1 and 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The display enclosure's real size, its clear lid and its wall thickness	The 4 screw holes, the gland hole and the sign face holes are drawn for about 200 x 140 x 24 mm; move them to suit	LDZ-DDR-003, P8
2	The e-paper driver board takes commands from the host FieldNode over a cable about 1 m long, and the pin assignment of the FieldNode sensor port	Sets the cable and connector between the sign and the FieldNode; the FieldNode port pinout is still open in FieldNode	LDZ-DDR-003, P8; FND-DEC-001
3	The band clamp range covers the band path for 60 to 90 mm poles (about 250 to 330 mm)	The band runs through the bracket as well as round the pole	LDZ-DDR-003, P7
4	The silicone cures against the primer used on the master and core	Some primers stop platinum silicone curing	Build plan, section 3.2
5	The potting's peak exotherm for about 175 cm ³ against the cells' temperature rating	The potting is poured over live cells	Build plan, safety stop S4
6	The magnetic offset of the cells' steel cans at the magnetometer, about 30 mm away	A fixed offset is learned at install; it must not use up the sensor range	LDZ-CAL-001, section 6
7	The road-marker epoxy's bond to sanded ASA	A second bond line beside the asphalt bond already at risk under R6	LDZ-DDR-003, A2

■ Value engineering

Value-engineering target: USD 120 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 116 for the two-puck kit (USD 4 under the target). The sign option (USD 101) and its host FieldNode (USD 126, priced in the FieldNode repo) are costed separately, and the one-off dome casting tooling (USD 35) is not part of the per-kit cost. Main cost drivers and savings worth trying:

- The largest lines, for two pucks, are the radio module and board (USD 28), the cell sets with diodes and pulse capacitor (USD 28), the magnetometer breakouts (USD 16), the base trays and potting (USD 12) and the domes (USD 10).
- Making the design constructable added USD 2 to the kit (sealant, screws and mould release in line 12) and USD 4 to the sign option (the channel brackets); the tooling is new.
- Savings worth trying: the magnetometer chip and the radio module on one small carrier board instead of two breakouts (a TRL 4 board design); cells bought in tens; one set of casting tooling shared by every puck made, so its cost falls per puck as more are cast.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: surface-bonded puck, magnetometer only, one puck per 7 m slot, two AA-size Li-SOCl ₂ cells, cast polyurethane dome over fully potted electronics, sign outside the kit cost, class C downlink to the sign, a feed that maps onto the Curb Data Specification	Amish: "i accept all your recommendations, go with them across all repos."	LDZ-DDR-001, LDZ-DDR-002
2026-09-25	TRL 3 items O2 to O7: gateway within 300 m (R4), road-marker epoxy bed and R6 restated, daylight-only sign (R11), private network server for the sign, diode per cell, 85 °C pulse capacitor, full potting and 6 mm crown radius, 11-byte payload if US915	Amish, same instruction	LDZ-DDR-002
2026-09-30	Make the design physically buildable as the build plan is drawn	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The resulting changes await his review (open decision 1)	LDZ-DDR-003
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering